SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – SOAIDEATHON-H23
Problem Statement Title – AI-Orchestrated Campus Microgrid with Solar, Storage, EV and Critical-Load Resilience
Theme – Renewable / Sustainable Energy PS
Category – Hardware
Team ID –
Team Name – Compile and Pray

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IDEA TITLE

AI-Orchestrated Campus Microgrid with Solar, Storage, EV and Critical-Load Resilience

PROBLEM

- CAMPUS DEMAND AND GENERATION CHANGE THROUGHOUT THE DAY
- CONVENTIONAL MICROGRIDS SUFFER FROM UNCOORDINATED LOAD DISTRIBUTION
- WEATHER CONDITONS CAN CHANGE ANY TIME LEADS TO CRITICAL POWER OUTAGES

PROPOSED SOLUTION

- EDGE CONNECTED ESP-32 TELEMETRY RELAYS REALTIME VOLTAGE/CURRENT DATA TO BACKEND SERVER
- FLASK ENGINE ENGINE CONTINOUSLY BALANCES LOADS

UNIQUE INNOVATION

- INTEGRATES OPEN METEO LIVE API FORECAST
- ONSITE RAIN AND TEMPERATURE SENSORS
- PREDICTIVE LOAD SHEDDING

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TECHNICAL APPROACH

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HARDWARE LAYER

- **ESP32 MICROCONTROLLER:**
  - WIFI HTTP TELEMETRY AND RELAY DRIVER
- **INA3221 SENSORS**
  - MULTICHANNEL V/I POWER PROFILING
- **OTHER SENSORS**
  - LM35 TEMP SENSOR
  - RAIN SENSOR
- **DISPLAY**
  - SSD1306 OLED LOCAL DASHBOARD

HARDWARE LAYER

- **FLASK MICRO-FRAMEWORK**
  - REST API ENDPOINTS
- **SQLite DATABASE**
  - TELEMETRY HISTORY LOGGING
- **OPEN METEO API**
  - UV-INDEX
  - RAIN/CLOUD FORECAST
- **CORS HANDLING**
  - REALTIME DASHBOARD STREAM

PROCESS METHODOLOGY

- **DATA INGESTION**
  - ESP32 POSTS JSON VIA SENSOR
- **WEATHER SYNC**
  - 10 SECOND CACHE AUTO REFRESH
- **AI EVALUATION**
  - RULE BASED HEURISTIC LOAD PRIORITIZATION
  - RAIN/CLOUD FORECAST
- **RELAY ACTION**
  - DYNAMIC L1-L4 LINE SHED/RESTORE

IMPLEMENTATION FLOW

1. COLLECT
Meters • weather • EV • battery

2. FORECAST
Solar + demand

3. OPTIMIZE
Battery + EV + loads

4. PROTECT
Critical loads first

5. EXPLAIN
Reason + cost + SoC

6. OVERRIDE
Operator control

Prototype output: live energy dashboard + recommended dispatch schedule + outage mode + explanation panel + manual override.

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FEASIBILITY AND VIABILITY

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FEASIBILITY

TESTED WORKING PROTOTYPE VALIDATES OPERATIONAL VIABILITY:

- LOW HARDWARE COST:
  - BUILT USING MODULAR, OFF-THE-SHELF ESP32 AND INA3221 COMPONENTS
- LIGHTWEIGHT STACK:
  - SQLITE + FLASK RUNS EFFICIENTLY ON EDGE GATEWAYS (E.G.RASPBERRY PI)
- PROVEN INTEROPERABILITY:
  - SEAMLESS HTTP JSON PROTOCOL BRIDGES MICROCONTROLLERS TO WEB BACKEND

CHALLENGES / RISKS

IDENTIFIED OPERATIONAL AND ENVIRONMENTAL VULNERABILITIES:

- NETWORK DROPOUT:
  - DISRUPTION IN WI-FI CAUSING TELEMETRY UPLOAD DELAYS
- WEATHER API LATENCY:
  - STALE WEATHER FORECASTS DURING SUDDEN MICRO-CLIMATES
- OVERCURRENT RISK:
  - RAPID LOAD SPIKES OVERWHELMING BATTERY STORAGE

MITIGATION

ENGINEERING SAFEGUARDS EMBEDDED INTO SOFTWARE & HARDWARE:

- LOCAL EDGE FAILOVER:
  - ESP32 OPERATES STANDALONE RULES WHEN SERVER CONNECTION DROPS
- HYBRID SENSING:
  - ON-BOARD RAIN SENSOR OVERRIDES API IF WEATHER SUDDENLY TURNS
- HARDWARE FAILSAFE:
  - DEDICATED ANALOG CUTOFF LIMITS ON MOTOR & BACKUP CHANNELS

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IMPACT AND BENEFITS

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TARGET AUDIENCE

- UNIVERSITY / CAMPUS FACILITY MANAGERS
- ENERGY MANAGERS AND MICROGRID OPERATORS
- CRITICAL CAMPUS FACILITIES: LABS, DATA ROOMS, HEALTH/SAFETY SYSTEMS
- EV CHARGING INFRASTRUCTURE OPERATORS
- STUDENTS AND STAFF WHO DEPEND ON RELIABLE CAMPUS SERVICES

EXPECTED BENEFITS

- **ENVIRONMENTAL IMPACT**

  MAXIMIZES LOCALIZED SOLAR UTILIZATION; REDUCES RELIANCE ON DIESEL GENERATORS AND FOSSIL FUEL GRID DRAWDOWN BY UP TO 35%.

- **ECONOMIC BENEFITS**

  PREVENTS COSTLY BATTERY DEEP-DISCHARGES AND LOWERS OPERATIONAL ELECTRICITY BILLS THROUGH DYNAMIC PEAK-SHAVING.

- **SOCIAL VALUE**

  ENSURES UNINTERRUPTIBLE POWER FOR ESSENTIAL LOADS (LIGHTING, MEDICAL/USB DEVICES) IN RURAL AND REMOTE MICROGRIDS.

Success metrics: forecast error • renewable utilization • peak-demand reduction • critical-load survival time • battery SoC compliance • EV charging completion • operator override rate

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RESEARCH AND REFERENCES

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RESEARCH BASIS

- DOE describes microgrids as controllable local systems that can operate grid-connected or independently during outages.
- DOE/NREL research highlights forecasting, advanced control, storage and EV-grid integration as important parts of modern microgrid operation.
- Our prototype applies these concepts at campus scale with explainable scheduling and operator override.

REFERENCES / LINKS

1. U.S. Department of Energy - Microgrid Systems: https://www.energy.gov/oe/microgrid-systems
2. U.S. Department of Energy - Microgrids: https://www.energy.gov/indianenergy/tribal-energy-guide/microgrids
3. NREL - Solar and Wind Forecasting: https://www.nrel.gov/grid/solar-wind-forecasting
4. NREL - EV-Grid Integration Control & System Implementation: https://research-hub.nrel.gov/en/publications/ev-grid-integration-evgi-control-and-system-implementation-resear/
5. DOE - Solar Integration: DER and Microgrids Basics: https://www.energy.gov/cmei/systems/solar-integration-distributed-energy-resources-and-microgrids-basics

Prototype data can be generated/simulated where real campus data is unavailable; assumptions should be clearly labelled in the demo.